

Day 4 Evidence Workbooklet — Instructor Key

Comparisons and Boundary Cases

Engineering practice → threshold rule → Python comparison → boundary evidence → support next step

Instructor key. Same layout as student copy; solutions filled in response boxes. Print format: duplex, left-stapled packet.

Name		Section		Date	
Score	____/44		Mastery check		secure / developing / needs support

Yesterday the pump-flow intake values were converted from text into numeric fields. Today the team uses those typed values to check a quality boundary: a flow reading is acceptable if it is inside a target band. The exact boundary rows matter because a reading exactly at the lower or upper limit may be allowed depending on the operator.

The problem today is comparison evidence: below, at, or above a limit; inclusive versus exclusive rules; and an exact sentence that explains a boundary case.

Engineering task	Check whether converted pump-flow readings meet an acceptable range before the team uses them in a go/no-go gate.
Computational move	Evaluate one comparison at a time and explain exact-boundary cases without jumping ahead to compound decisions.
Python focus	<, <=, >, >=, ==, !=, comparison result, inclusive limit, exclusive limit.
Evidence to keep	reading value, boundary value, operator, true/false result, and a sentence explaining whether the boundary is included.

Day 4 boundary rules

1. A comparison expression produces **True** or **False**.
2. <= and >= include equality at the boundary.
3. < and > do not include equality at the boundary.
4. == checks equality; it does not assign a value.
5. A reliable threshold note names the value, the limit, the operator, and the result.

1 Read the threshold notebook one comparison at a time

recognize predict

Use the converted numeric fields from the intake record. Do not combine the comparisons yet; Day 5 will do that.

```
1 # Converted numeric fields from the intake record
2 target_flow_lpm = 20.0
3 low_limit_lpm = 19.6
4 high_limit_lpm = 20.4
5 reading_low_edge = 19.6
6 reading_middle = 20.0
7 reading_high_edge = 20.4
8 reading_high_miss = 20.5
9
10 # Individual comparison checks
11 reading_low_edge >= low_limit_lpm
12 reading_low_edge > low_limit_lpm
13 reading_high_edge <= high_limit_lpm
14 reading_high_edge < high_limit_lpm
15 reading_high_miss <= high_limit_lpm
16 reading_middle == target_flow_lpm
```

Pts.	Prompt	My evidence
1	Which variables define the lower and upper allowed flow limits?	Key: low_limit_lpm and high_limit_lpm.
1	Which reading is exactly equal to the lower limit?	Key: reading_low_edge = 19.6.
1	Which reading is exactly equal to the upper limit?	Key: reading_high_edge = 20.4.
1	Which reading is above the upper limit?	Key: reading_high_miss = 20.5.

2 Predict comparison results at exact boundaries

predict explain

Complete the table. Explain the result using the operator, not just the number.

Pts.	Comparison expression	True or False?	Boundary reason
2	<code>reading_low_edge >= low_limit_lpm</code>	Key: True	Key: Equal to lower limit; >= includes equality.
2	<code>reading_low_edge > low_limit_lpm</code>	Key: False	Key: Equal is not greater than.
2	<code>reading_high_edge <= high_limit_lpm</code>	Key: True	Key: Equal to upper limit; <= includes equality.
2	<code>reading_high_edge < high_limit_lpm</code>	Key: False	Key: Equal is not less than.

Common trap to check

A reading exactly at the limit is included by `<=` or `>=`, but it is not included by `<` or `>`. The operator decides the boundary, not the word "limit" by itself.

3 Separate equality from assignment and mismatch checks

recognize

predict

Python uses == for equality checks and = for assignment. This page keeps comparison evidence separate from stored state.

Pts.	Expression or line	Result or effect	What evidence does it create?
2	reading_middle == target_flow_lpm	Key: True	Key: Equality comparison; no assignment.
2	reading_high_miss != target_flow_lpm	Key: True	Key: 20.5 is not equal to 20.0.
2	reading_middle = target_flow_lpm	Key: assignment	Key: Stores 20.0 in reading_middle; not a comparison.
2	reading_high_miss <= high_limit_lpm	Key: False	Key: 20.5 is above 20.4.

Pts.	Prompt	My response
2	Why is == a safer symbol than the word "is" when writing comparison evidence?	Key: == names an equality comparison explicitly; "is" can be ambiguous in prose.
2	Why should a threshold note not say only "near the target"?	Key: It must name the value, limit, operator, and whether equality is included.

4 Build a boundary table before combining conditions

[trace](#) [explain](#)

Complete the boundary table. The last column should be a short engineering sentence, not only **True** or **False**.

Pts.	Reading	Lower check: ≥ 19.6	Upper check: ≤ 20.4	Boundary evidence sentence
2	19.5	Key: False	Key: True	Key: Below lower limit; not allowed.
2	19.6	Key: True	Key: True	Key: At lower limit; inclusive lower boundary allows it.
2	20.4	Key: True	Key: True	Key: At upper limit; inclusive upper boundary allows it.
2	20.5	Key: True	Key: False	Key: Above upper limit; not allowed.

Bridge to Day 5

Tomorrow the team will combine separate comparison results into a go/no-go expression. Today the important habit is to make each comparison traceable first: value, operator, limit, result, and boundary sentence.

5 Debug a boundary claim

[debug](#) [test](#) [explain](#)

A teammate writes this note after checking the lower edge reading:

Boundary note to debug

The 19.6 L/min reading fails because it is not greater than the lower limit of 19.6 L/min.

Pts.	Prompt	My response
2	What part of the teammate's sentence uses a real comparison result?	Key: 19.6 > 19.6 is False.
2	What rule did the teammate use that may not match the allowed boundary?	Key: They used an exclusive > rule. If the allowed boundary is inclusive, use >=.
2	Write the inclusive comparison that would allow the exact lower-limit reading.	Key: reading_low_edge >= low_limit_lpm.
2	Rewrite the boundary note so it names the operator and explains whether the exact limit is included.	Key: Using >=, 19.6 >= 19.6 is True, so the exact lower-limit reading is allowed.

Boundary-test habit

When a threshold rule matters, test at least three rows: just below the boundary, exactly at the boundary, and just above the boundary. Exact-boundary rows catch many comparison mistakes.

6 Mastery check and support next step

[transfer](#)[explain](#)

Use your own evidence from this workbooklet. Do not write a generic answer.

Pts.	Prompt	My response
2	Give one example where changing from > to >= changes the result at an exact boundary.	Key: <code>reading_low_edge > low_limit_lpm</code> is False, but <code>reading_low_edge >= low_limit_lpm</code> is True.
2	In one or two sentences, explain why Day 3 type/cast evidence is needed before Day 4 boundary checks.	Key: Boundary comparisons require numeric values. Day 3 casting makes sure text fields have become floats before comparison.

My mastery check

Mark the strongest statement that is true after this workbooklet.

☐	Secure: I can evaluate comparison operators and explain exact lower or upper boundary cases.
☐	Developing: I can evaluate some comparisons, but I still need support with inclusive versus exclusive limits.
☐	Needs support: I need a smaller example before I can trust my boundary-check evidence.

My next support move

My issue is mostly: setup / Python syntax / tracing / operator choice / boundary cases / confidence / time.

The first boundary where I got uncertain was: _____

My next small action is: ask a partner / ask instructor / rebuild the boundary table / test just-below-at-above rows / try the recovery version.

Workbooklet target: I can evaluate Python comparison operators, explain exact-boundary cases, and prepare separate comparison evidence before Boolean combinations.

Copyright © 2026 Lance L.A. White. All rights reserved.

VIP Lab Alignment

Bridge Day 4 • Contract 2d4127aac856

This page adds a current lab handoff while preserving the workbook's independence-ramp tasks.

Why this page is here

VIP Lab Day(s): 4

Activities: 18.13, 18.15, 18.16, 18.17

Carry below-at-above boundary evidence into complete comparison and ordered-branch programs without requiring the supplied-loop activity yet.

Open these current notebook cells

Activity / stable cells	Notebook work	Evidence to carry forward
18.13 18-13-work / 18-13-transfer / 18-13-cold	Compare With Tolerance	Compute absolute distance between two values.
18.15 18-15-work / 18-15-transfer / 18-15-cold	Lamination Fee	Translate three quantity intervals into ordered branches.
18.16 18-16-work / 18-16-transfer / 18-16-cold	Temperature Status	Compare a measurement with a maximum.
18.17 18-17-work / 18-17-transfer / 18-17-cold	Pickup Window	Calculate round-trip travel time.

Readiness check before you code

1. Choose one mapped activity and write the below, exact-boundary, and above test values before coding.

Answer and explanation: Credit a valid three-case boundary set tied to the activity, including tolerance equality, page thresholds, maximum plus five, or the travel-time boundary.

2. For the selected activity, list input conversion, named comparison evidence, first matching branch, and exact formatted output.

Answer and explanation: The chain must begin with numeric conversion, preserve the comparison result, respect branch priority, and end with the specified label and formatting.

Notebook and submission procedure

1. Use the sample and transfer cells for formative practice; attempt before opening a reveal.
2. Complete the end-of-notebook Independent Program Studio without a reveal or copied solution. Only the Studio cells listed above are scored for independent mastery in the matching Lab Day notebook.
3. Save or download the completed notebook as one `.ipynb` file.
4. Upload that one notebook to the matching Gradescope assignment. Do not export a `.py` file or create a ZIP.